

Elicit vs. Teach, Mechanistically: Eleven Metrics on the TinyStories-1B Twins

MARS V — Mechanistic Understanding of Elicitation vs. Teaching

Working results document, September 2026

Setup

The experiment. Two identical copies (“twins”) of a 1B-parameter language model pretrained only on children’s stories — no arithmetic anywhere in the corpus — are fine-tuned on the same target task: natural-language addition and subtraction of 1–4-digit numbers, **What is the difference between 8 and 5784? $\rightarrow$ -5776.** The training protocol is identical for both. The *only* difference is the checkpoint the fine-tune starts from:

- **Teach twin:** the blank base model. The capability is absent.
- **Elicit twin** (“TS1B-latent”): the blank base plus two installations, neither of which shows the target task: (i) arithmetic in symbols only ($23 + 45 = 68$, 1M examples, full fine-tuning) and (ii) a word $\leftrightarrow$ symbol binding dose that only *rewrites* questions (**What is the sum of 621 and 5068? $\leftrightarrow$ 621 + 5068**) and never shows an answer. This twin can compute in symbols and knows what the words mean, but on the target task it scores 0.000: asked a question, it rewrites it instead of answering. The capability is latent.

Every construction step has a control (Table 1); the two twins are behaviourally identical on the target task before fine-tuning.

Construction step / check	value	control
symbol install: exact match on $23 + 45 =$	0.67–0.73	blank twin: 0.00
corpus census: “sum” in 439M words of stories	$21\times$	“equals” 17, “minus” 57
binding dose: question $\rightarrow$ symbols rewrite exact	0.484	blank + same dose: 0.000
sum-only dose on <i>difference</i> questions	writes “+”	binding is per word
exact match on the target task, before fine-tuning	0.000	= blank twin

Table 1: The elicit twin before fine-tuning: can compute in symbols, understands the words, still answers nothing. Everything below is about what differs inside the two twins and what that difference does.

Endpoints. Seven fine-tuned models enter the results (Table 2). With the paper’s adapter protocol (LoRA), the teach twin stops by the protocol’s own convergence rule far below the elicit twin — 0.093 with 1M examples and 0.139 with four times as many unique examples — which is where the paper’s own Figure 2 puts this model (1–2 nats per token at 1M; “stalls at lower accuracy”). Teaching this base needs full fine-tuning. Two fully fine-tuned teach endpoints are carried through every metric: the *format-installed* twin (the paper’s “pre-teach format” control: the blank base first fine-tuned on a random-label dose in the target’s own format, so it already emits a number after every question; exact match 0.000) fine-tuned to convergence, which reaches 0.424; and the

blank twin fine-tuned to convergence, which reaches 0.819. The elicit twin was fully fine-tuned with the identical recipe (0.986). The format-installed pair is the design-matched comparison — both parents are the blank base plus a full-FT dose, both already output numbers, only one has the algorithm — and the blank-base pair is the better-performing one; the two taught endpoints agree on every metric below. LoRA endpoints are kept as robustness rows; two metrics need adapter snapshots (formation time, weight travel) and exist only for the LoRA runs.

endpoint	parent	method	data	steps (passes)	stop	exact match
elicit LoRA	TS1B-latent	LoRA $r=512$	1M	11,500 (1.5)	converged	0.981
teach LoRA 1M	blank	LoRA $r=512$	1M	25,000 (3.2)	converged	0.093
teach LoRA 4M	blank	LoRA $r=512$	4M	39,500 (1.3)	converged	0.139
teach FT	blank	full FT	4M	62,500 (2.0)	ceiling	0.771
teach FT, converged	blank	full FT	4M	72,000 (2.3)	converged	0.819
teach FT, format-installed	format-installed	full FT	4M	58,500 (1.9)	converged	0.424
elicit FT	TS1B-latent	full FT	4M	21,500 (0.7)	converged	0.986

Table 2: Fine-tuned endpoints. LoRA learning rate 3.53×10^{-4} , full FT 2×10^{-5} ; batch 128; seed 316; training stops when validation loss stops improving (eps/k rule). Exact match = 0-shot accuracy on 1,024 held-out problems. The 4M file is the 1M file plus 3M new unique problems, disjoint from every evaluation set. “Converged” teach FT continues the 2-pass run from its checkpoint until the rule fires (optimizer state restarted).

Two terms used throughout. *Answer position*: the last prompt token, where the model predicts the first token of the answer; every activation measurement is taken there unless stated. *Residual stream*: the model’s running internal state at a token position, a 2048-dimensional vector that every layer reads and adds to. Everything else is defined where it is used.

How to read this document. Eleven metrics, each with the same three blocks: **Elicit vs. teach, in one sentence** (what the metric asks and which way each regime should fall), **How it is measured** (the procedure, complete enough to reproduce), **The numbers** (the headline value for each regime; every value is in the table or figure it points to). The metrics are grouped by what they look at: the *cost* of learning (M1), the *machinery* that results (M2–M4), the *dynamics* of building it (M5–M8), and the *latent state* before any target training (M9–M11). Table 11 at the end puts every metric for the full-FT endpoints on one page.

A. Cost of learning

M1 Learning cost — *cheaper with every example vs. the hump*

Elicit vs. teach, in one sentence. Does each new example cost less than the last, because it only connects a capability that exists (elicit), or does the cost first rise, because structure is being built (teach)?

How it is measured. Prequential excess description length (EDL), the paper’s learning-cost measure. Train on a stream of n examples; before each example is trained on, record the extra nats the model needs to predict its answer tokens beyond a converged reference; sum over the stream and divide by the number of answer tokens (EDL per token). Repeat on a sweep $n = 100 \dots 4 \times 10^6$ with the identical LoRA protocol (19 rungs per arm, all converged), plus 0-shot exact match on 1,024 held-out problems at each rung. A falling curve means each example costs less than the last; a down–up–down curve (“the hump”) means mid-run examples are the most informative because structure is being built. Three arms: the elicit

twin, the blank teach twin, and the format-installed teach twin. The full-FT endpoints have no sweep; their cost is reported as steps to convergence and held-out test loss.

The numbers. Figure 1, Table 3. EDL/token falls monotonically for the elicit twin ($4.53 \rightarrow 0.09$) and goes down-up-down for both teach twins (blank: minimum 0.78, peak 2.02 at 215K; format-installed: minimum 0.64, peak 1.33 at 100K). Exact match at 1M: 0.98 vs 0.09 / 0.08; under full FT 0.986 vs 0.424 (format-installed) / 0.819 (blank), in 0.7 vs 1.9–2.3 passes. *Reading:* eliciting connects a capability that exists; teaching builds one, and the hump survives removing the format, so it is the algorithm being built.

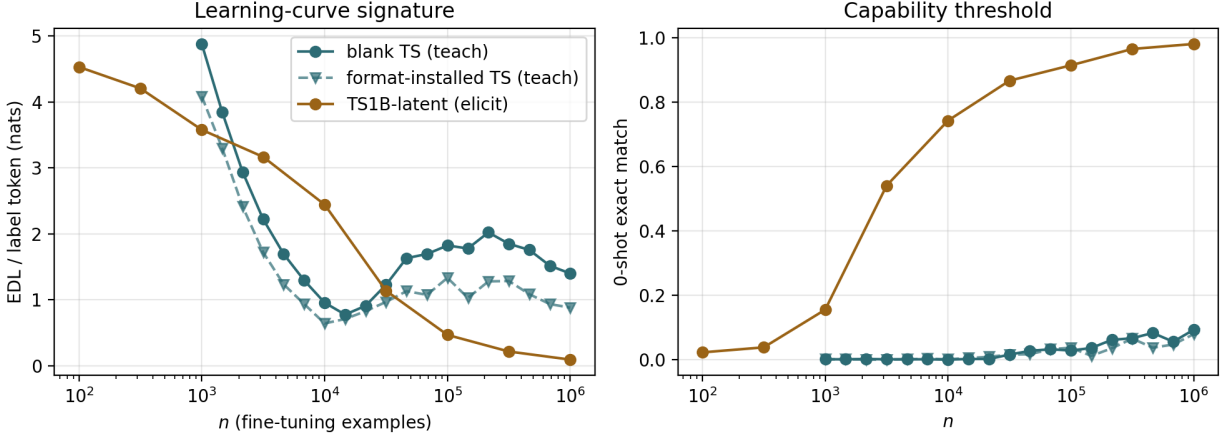


Figure 1: EDL/token (left) and 0-shot exact match (right) vs n ; teal = teach twin from the blank base, dashed teal = teach twin from the format-installed base, gold = elicit twin.

n	elicit (TS1B-latent)		teach (blank)		teach (format-installed)	
	EDL/tok	EM	EDL/tok	EM	EDL/tok	EM
100	4.533	0.023	6.511	0.007	—	—
316	4.208	0.038	6.387	0.033	—	—
1,000	3.579	0.155	4.882	0.001	4.087	0.001
3,162	3.168	0.541	2.221	0.001	1.720	0.000
10,000	2.444	0.743	0.956	0.000	<i>0.641</i> (min)	0.001
14,678	—	—	<i>0.775</i> (min)	0.002	0.708	0.005
31,623	1.137	0.867	1.228	0.016	0.968	0.016
100,000	0.467	0.915	1.824	0.028	<i>1.333</i> (peak)	0.037
215,443	—	—	<i>2.022</i> (peak)	0.062	1.279	0.035
316,228	0.215	0.966	1.849	0.066	1.285	0.065
1,000,000	0.092	0.981	1.399	0.093	0.880	0.079
4,000,000	—	—	0.930	0.139	—	—

Table 3: Per-rung values, LoRA protocol (EDL in nats per answer token; EM = exact match; 19 rungs per teach arm, all converged; rungs shared with the elicit sweep shown plus each arm’s minimum and peak).

B. The machinery that results

M2 Circuit overlap — *the same circuit vs. a new circuit*

Elicit vs. teach, in one sentence. Does the fine-tuned model solve the task with the circuit its parent already had (elicit) or with a circuit that did not exist before (teach)?

How it is measured. *Building a circuit.* The model is split into 528 nodes: 512 attention heads (32 per layer $\times$ 16 layers; heads move information between token positions) and 16 MLP blocks (one per layer; they transform it in place). For each node, attribution patching scores how much the answer’s logit would change if that node’s activation at the answer position were replaced by its value on a different problem: gradient of the answer logit-difference with respect to the node’s activation, times the activation difference between the clean and the corrupted problem, averaged over 128–256 problem pairs. The 32 highest-scoring nodes are the circuit. A model that cannot do the task gives a noise map, which the tool flags (its answer logit-difference is ≈ 0). *Comparing circuits.* Jaccard@32 = shared nodes $\div$ union of the two 32-node sets. Two random circuits overlap at 0.031. The *ceiling* is the overlap of two circuits of the *same* model computed from two disjoint halves of the data — how repeatable a circuit is (0.42–0.68 here); an overlap at the ceiling means “the same circuit”. Spearman = rank correlation of the node scores over the union. Also the layer histogram of each circuit’s top-16 nodes, and for the LoRA-1M pair the change at the level of connections between nodes (edge-level $\Delta S = 1 - \text{Jaccard}$ over the 256 strongest node $\rightarrow$ block edges).

The numbers. Table 4, Figure 2. The two elicited children (LoRA and full FT) share one circuit: 0.641, at the ceiling. Elicited vs taught: 0.306 for the design-matched full-FT pair (ceilings 0.56 / 0.52; score correlation 0.03), 0.23–0.39 for the other pairs. The elicited circuit is the parent’s own symbol circuit re-weighted (0.46–0.52, top four nodes identical); the blank parent has no circuit to inherit. Shared by all: the late-layer MLPs; different: how the operands are fetched. *Reading:* elicitation re-weights the circuit the parent already had; teaching builds a related but different one.

Comparison	Jaccard@32	chance	ceiling(s)
parent’s symbol circuit $\leftrightarrow$ elicited child (two eval surfaces)	0.455–0.488 / 0.524	0.031	0.684
elicited child @3K $\leftrightarrow$ @1M	0.391	0.031	0.684
elicited child, full FT $\leftrightarrow$ elicited child, LoRA	0.641 (Spearman 0.54)	0.031	0.561 / 0.684
blank parent $\leftrightarrow$ taught child	undefined (parent map is noise)	–	–
elicited FT $\leftrightarrow$ taught FT, format-installed	0.306 (Spearman 0.03)	0.031	0.561 / 0.524
elicited FT $\leftrightarrow$ taught FT, blank, 2 passes / converged	0.333 (Spearman 0.16) / 0.333 (0.28)	0.031	0.561 / 0.422
taught FT, converged $\leftrightarrow$ taught FT before continuation	0.600 (Spearman 0.57)	0.031	0.422 / 0.422
taught FT, format-installed $\leftrightarrow$ taught FT, blank, converged	0.362 (Spearman 0.18)	0.031	0.524 / 0.422
elicited $\leftrightarrow$ taught, LoRA pair (4M)	0.391 (Spearman 0.28)	0.031	0.684 / 0.524
elicited $\leftrightarrow$ taught, LoRA pair (1M)	0.231 (Spearman –0.11)	0.031	0.684 / 0.561
taught, full FT $\leftrightarrow$ taught, LoRA 4M	0.391 (Spearman 0.11)	0.031	0.422 / 0.524

Table 4: Circuit overlaps. All maps pass the performing-regime check.

M3 Circuit faithfulness — *does the circuit carry the computation?*

Elicit vs. teach, in one sentence. Do the 32 nodes called “the circuit” really produce the answer — in both regimes — so that every overlap in M2 compares two real mechanisms?

How it is measured. True activation patching, in two directions. *Sufficiency:* run a corrupted problem, copy the clean problem’s activations into only the top- k nodes, and measure

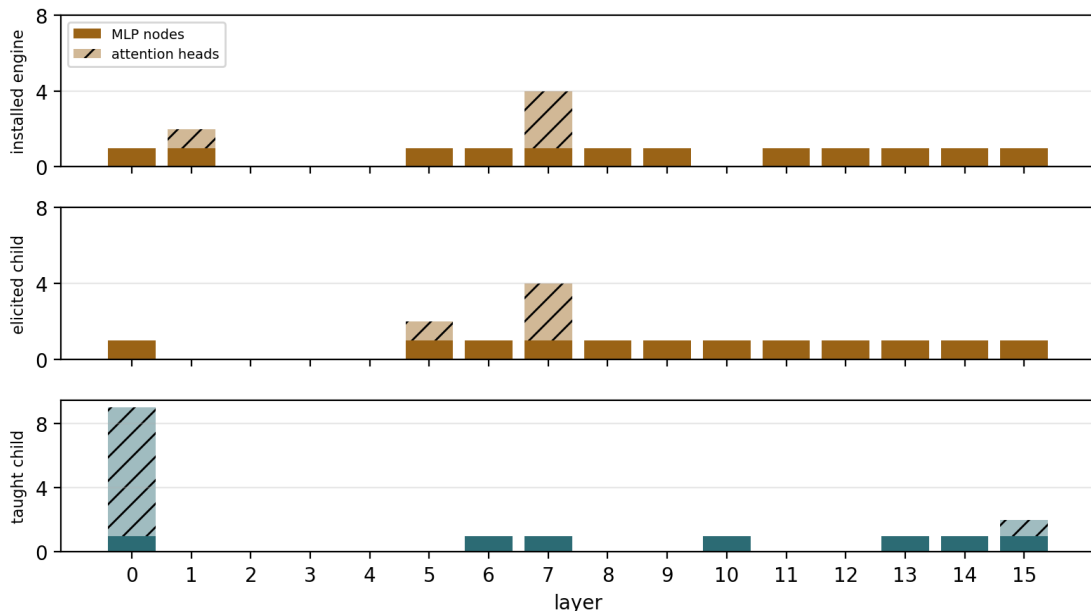


Figure 2: Top-16 node layer histograms (solid = MLP, hatched = attention) for the parent’s symbol circuit, the LoRA-elicited child and the LoRA-1M taught child.

what fraction of the clean-vs-corrupt answer logit-difference is recovered. *Necessity*: run the clean problem, corrupt only the top- k nodes, and measure what fraction is destroyed. 128 problem pairs, $k = 8, 16, 32, 64, 128, 528$; 1.0 means the top- k nodes alone account for the whole behaviour.

The numbers. Top-8 of 528 nodes recover / destroy the answer at 0.92–1.00 for every model (elicit FT 0.923 / 0.943, teach FT format-installed 0.983 / 0.988); top-32 ≥ 0.996 everywhere (Table 11). *Reading*: no difference — both regimes build compact, load-bearing circuits, so M2, M4 and M5 compare two real mechanisms.

M4 Head roles — *the same reader heads vs. rediscovered ones*

Elicit vs. teach, in one sentence. Does the fine-tuned model keep the parent’s heads that read the two numbers (elicit) or have to find reader heads of its own (teach)?

How it is measured. Desiderata Component Masking (Prakash et al. 2024). For one piece of information at a time, take a problem and a counterfactual problem that differs only in that piece, and learn (by gradient descent on a mask over heads, with a sparsity penalty) the smallest set of attention heads such that copying those heads’ activations from the counterfactual run into the clean run makes the model answer the counterfactual problem. That set is the heads carrying that piece (its “role”). The set is validated by its counterfactual-flip accuracy against the ceiling obtained by copying every head; a set is “compact” only if it reaches that ceiling. Role sets are compared across models by Jaccard (chance ≈ 0.05); 64–128 counterfactual pairs, heads only.

The numbers. Table 5. Elicitation keeps the parent’s operand-reading heads: 0.88 / 0.90 overlap (first / second operand), flip accuracy at the ceiling. Every performing taught model rediscovers most of the same heads in a larger set: 0.88 / 0.66 (format-installed full FT) and

0.76 / 0.66 (blank full FT) vs the parent’s engine. The LoRA-taught children that never learned the task have no compact set at all (flip accuracy ≤ 0.04). *Reading:* operand reading is a layer-0 function of the substrate; elicitation keeps it intact, teaching has to find it again — a graded difference, not an absence.

Comparison (first operand / second operand)	$ A $	$ B $	Jaccard	flip acc. / ceiling
parent $\leftrightarrow$ elicited (LoRA), symbol surface	30 / 27	32 / 30	0.879 / 0.900	at ceiling
parent $\leftrightarrow$ elicited (LoRA), operation	25	24	0.750	at ceiling
elicited (full FT) $\leftrightarrow$ elicited (LoRA)	31 / 25	36 / 30	0.811 / 0.833	0.98 / 0.98
elicited (full FT) $\leftrightarrow$ parent	31 / 25	30 / 27	0.694 / 0.857	–
taught (full FT) $\leftrightarrow$ elicited (LoRA)	37 / 37	36 / 30	0.587 / 0.763	0.81 / 0.79 (0.81 / 0.80)
taught (full FT) $\leftrightarrow$ elicited (full FT)	37 / 37	31 / 25	0.700 / 0.632	–
taught (full FT) $\leftrightarrow$ parent	37 / 37	30 / 27	0.763 / 0.684	–
taught (FT, format-installed) $\leftrightarrow$ parent	32 / 41	30 / 27	0.879 / 0.659	0.59 / 0.54 (0.59 / 0.55)
taught (FT, format-installed) $\leftrightarrow$ elicited (LoRA)	32 / 41	36 / 30	0.659 / 0.690	–
taught (FT, blank, converged) $\leftrightarrow$ parent	37 / 36	30 / 27	0.763 / 0.658	0.85 / 0.81 (0.85 / 0.83)
taught (FT, blank, converged) $\leftrightarrow$ elicited (LoRA)	37 / 36	36 / 30	0.587 / 0.737	–
taught (LoRA 4M) $\leftrightarrow$ elicited (LoRA)	12 / 21	36 / 30	0.043 / 0.041	0.01 / 0.03 (0.20 / 0.14)
taught (LoRA 1M) $\leftrightarrow$ elicited (LoRA)	9 / 21	36 / 30	0.125 / 0.085	0.02 / 0.04 (0.15)

Table 5: Head roles by Desiderata Component Masking. $|A|$, $|B|$ = size of the two role sets; flip accuracy is reported for set A .

C. Dynamics: how the machinery comes into being

M5 Circuit formation time — *present at step 1 vs. assembled mid-run*

Elicit vs. teach, in one sentence. Is the final circuit present from the first step (elicit) or assembled somewhere in the middle of training (teach)?

How it is measured. Adapter snapshots were saved along the LoRA runs (128 per run, log-spaced). For each snapshot, build the circuit exactly as in M2 and take its Jaccard@32 with the run’s final circuit; read against chance (0.031) and the split-half ceiling. The step at which the curve leaves the noise floor is when the circuit starts to exist; the step at which it reaches the ceiling is when it stops changing.

The numbers. Figure 3. Elicit: 0.49 overlap with the final circuit at step 1 ($16\times$ chance), locked from step 1.5K of 11.5K. Teach (LoRA 1M): at the noise floor until step 1.3K, then built over steps 1.3K–5.4K, where its EDL peaks (M1); LoRA 4M keeps churning to the end. *Reading:* the elicited circuit is present before training starts; the taught circuit is assembled mid-run, exactly while the learning cost is rising.

M6 Gradient pressure — *a dying gradient vs. a growing one*

Elicit vs. teach, in one sentence. Does the gradient fade as training proceeds because there is nothing left to change (elicit), or grow because structure is still being built (teach)?

How it is measured. The global gradient norm before clipping, logged at every update. Reported: its peak and the step of the peak; its mean over the first 1%, the middle, and the last 10% of steps; the ratio first-1% $\div$ last-10% (above 1 = decays, below 1 = grows); and the cumulative gradient mass $\sum_t \|g_t\|$. For the LoRA runs also how that mass splits across attention query/key weights (QK), attention value/output weights (VO) and MLP weights;

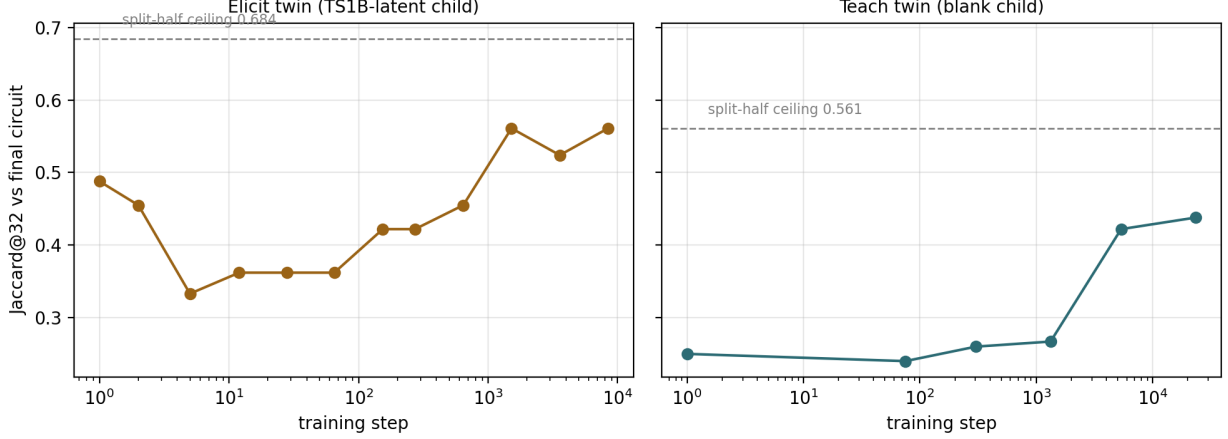


Figure 3: Circuit formation, elicit (left) vs teach (right); dashed = split-half ceiling.

the full-FT trainer logs the global norm only. Norms are comparable within a method, not across methods (different parameter sets).

The numbers. Table 6, Figure 4. Elicit’s gradient decays by 3.3–3.4 $\times$ from the first 1% of steps to the last 10%, under either method. Teach’s grows: 40–50 $\times$ under LoRA, 1.4–1.5 $\times$ under full FT. Cumulative gradient mass, full FT: 44K (elicit) vs 378K (format-installed teach) and 353K (blank teach), 8–9 $\times$; LoRA: 150–280 $\times$, aimed at attention (VO share 0.87–0.93). *Reading:* elicitation converges and its gradient dies; teaching keeps generating pressure while the structure is built.

	elicit LoRA 1M	teach LoRA 1M	teach LoRA 4M	elicit FT	teach FT, blank (+ cont.)	teach FT, format-inst.
steps	11,500	25,000	39,500	21,500	62,500 (+ 9,500)	58,500
peak gradient norm (step)	0.441 (96)	17.4 (23,657)	22.0 (36,931)	34.0 (1)	32.0 (3,844)	70.9 (50,860)
mean norm: first 1% / middle / last 10%	0.183 / 0.060 / 0.056	0.169 / 5.21 / 6.80	0.160 / 6.61 / 7.78	4.80 / 1.85 / 1.39	2.30 / 5.31 / 3.43	2.69 / 4.86 / 3.77
first 1% $\div$ last 10%	$\times 3.3$ (decays)	$\times 0.02$ (grows $\sim 40\times$)	$\times 0.02$ (grows $\sim 50\times$)	$\times 3.4$ (decays)	$\times 0.7$ (grows 1.5 $\times$)	$\times 0.7$ (grows 1.4 $\times$)
cumulative gradient mass $\sum_t \ g_t\ $	783	116,877	220,685	44,021	353,277 (+ 30,954)	378,170
gradient mass per step	0.068	4.68	5.59	2.05	5.65 (3.26)	6.46
mass share QK / VO / MLP	.18 / .55 / .27	.12 / .87 / .01	.05 / .93 / .01	–	–	–

Table 6: Raw gradient strength (pre-clip global norm at every update). LoRA and full-FT norms are on different parameter sets and are compared within a method only.

M7 Weight write — a small update vs. a large one

Elicit vs. teach, in one sentence. Is the change to the weights small and efficient (elicit) or large and expensive per unit of capability gained (teach)?

How it is measured. Relative weight change $\|\Delta W\|/\|W\|$ between child and parent — exact from the LoRA factors ($\Delta W = BA$), or from the checkpoint difference for full fine-tuning — over all attention and MLP matrices, with the effective rank of ΔW (participation ratio of its spectrum; 2048 = full rank). Along the LoRA runs, from the adapter snapshots: total travel $\|\Delta W_t\|_F$ vs step, its peak and final speed per step, and capability per unit travel = final exact match $\div$ total travel.

The numbers. Table 7. Relative weight write, full FT: 0.050 (elicit) vs 0.086 (format-installed teach) and 0.102 (blank teach), 1.7–2.0 $\times$, at the same effective rank; LoRA: 0.095 vs 0.21–0.26, 2.2–2.8 $\times$. LoRA travel 120 vs 457, and capability per unit travel 8.1×10^{-3} vs

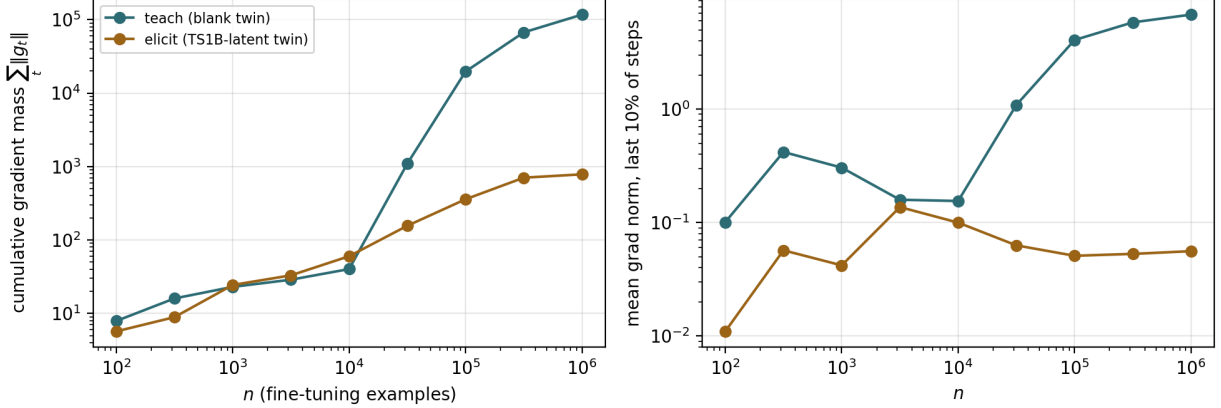


Figure 4: Cumulative gradient mass (left) and late-run gradient norm (right) vs n ; teal = teach, gold = elicit.

2.0×10^{-4} (40 $\times$). *Reading*: teaching writes two to four times more into the weights and gets forty times less capability per unit written.

	elicit (TS1B-latent)	teach (blank)
relative write $\ \Delta W\ /\ W\ $, LoRA 1M	0.095	0.212
relative write, LoRA 4M	—	0.264
relative write, full FT (4M): blank 2 passes / converged / format-installed	0.050	0.094 / 0.102 / 0.086
effective rank of ΔW , full FT (of 2048)	554	571 / 580 / 481
steps to convergence (LoRA 1M)	11,500	23,496
peak / final writing speed per step (LoRA 1M)	0.218 / 0.018	0.306 / 0.022
total weight travel $\ \Delta W\ _F$ (LoRA 1M)	120	457
final exact match (LoRA 1M)	0.981	0.093
capability per unit travel (LoRA 1M)	8.1×10^{-3}	2.0×10^{-4}

Table 7: Weight-space write, elicit vs teach (travel curves in 05_weights/fig_weight_travel).

M8 State-shift direction — a different vector per problem vs. one shared vector

Elicit vs. teach, in one sentence. Does fine-tuning change the model’s internal state by a different vector for every problem — the answer, routed to the output (elicit) — or by the same push on every problem — moving the model into a new mode (teach)?

How it is measured. Run the same 256 problems through parent and child (both in fp32) and take the difference of their residual streams at the answer position, layer by layer. *Size*: its norm relative to the parent’s state. *Direction*: stack the 256 difference vectors and compute the fraction of their energy in the single best direction (PC1; 1.0 means every problem’s difference is the same vector). The reference for “unrelated vectors” is the PC1 of the parent’s own 256 states, given alongside. The same PC1 is recomputed after projecting out the directions that encode the answer token and the parent’s top-1 token, so that “the same vector” cannot simply be the answer. Also the cosine of each shift to the mean shift, and, as a functional counterpart, $\text{KL}(\text{parent} \parallel \text{child})$ at the answer position vs on generic story text.

The numbers. Figure 5, Table 8. PC1 of the shift at the read-out: elicit 0.056 (full FT) and 0.063 (LoRA), below the parent’s own states (0.23) — no shared direction; teach 0.567

(format-installed full FT), 0.526 (blank full FT), 0.528 (LoRA 4M), unchanged after removing the answer content (0.56 / 0.53 / 0.52). The child with no format left to learn has the largest shared component, so the component is the algorithm’s mode, not the format. Size follows the method, not the regime. *Reading*: elicitation adds a problem-specific vector to a state that is already problem-specific; teaching moves every problem by the same push into answering mode first.

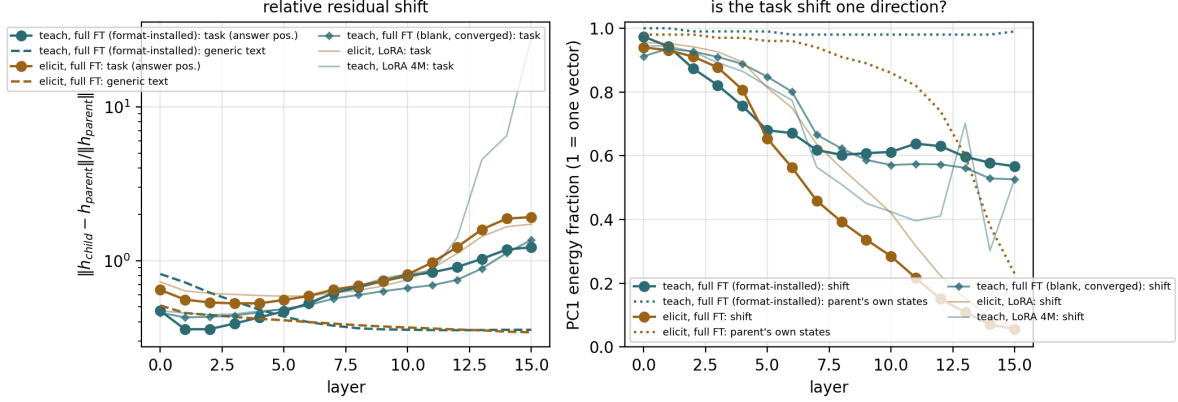


Figure 5: Full-FT endpoints (thick: elicit vs format-installed teach; diamonds: blank-base teach converged) with the LoRA endpoints thin. Left: relative residual shift by layer at the answer position (solid) and on generic story text (dashed). Right: PC1 energy fraction of the 256 per-problem shift vectors, with the parent’s own states dotted.

	elicit LoRA 1M	teach LoRA 4M	elicit FT	teach FT, blank (conv.)	teach FT, format-inst.
final exact match	0.981	0.139	0.986	0.819	0.424
PC1 of shift at the read-out (parent’s own states)	0.063 (0.23)	0.528 (0.98)	0.056 (0.23)	0.526 (0.98)	0.567 (0.99)
PC1 after removing answer-token directions	0.072	0.520	0.061	0.528	0.561
cosine of each shift to the mean shift	0.26	0.26	0.23	0.74	0.80
shift at answer position, final layer (relative)	1.72	28.6	1.92	1.36	1.22
KL(parent child) at the answer position (nats)	14.3	14.0	16.4	22.0	12.3
KL(parent child) on stories (nats/token)	0.067	0.326	0.721	1.790	1.568

Table 8: Residual-stream shift parent→child at the answer position, 256 problems, fp32. Blank-base teach FT before continuation: PC1 0.521, final-layer shift 1.42; LoRA-1M teach: PC1 0.351, shift 14.1.

D. The latent state: what it means and what it permits before training

M9 Latent reach — do the words already reach the machinery?

Elicit vs. teach, in one sentence. Before any target training, does a natural-language question already put the arithmetic components into a problem-specific state (elicit), or is there nothing for the words to reach (teach)?

How it is measured. Render each problem two ways — What is the sum of 621 and 5068? and 621 + 5068 = — and record every node’s activation at the answer position. Per node: the mean cosine between the two renderings of the *same* problem minus the mean cosine between renderings of *different* problems (the “bridging index”). Positive means the words put

the node into the state the symbols put it in. MLP blocks only: attention heads share digit tokens across renderings and would score trivially. Compared between the symbol-only engine (no binding dose) and the elicit parent (binding dose installed). A second, geometry-only check: how problem-specific the answer-position state itself is, as PC1 and cosine-to-mean across the 256 problems (≈ 1 means the same state for every problem).

The numbers. Bridging index of the arithmetic MLPs on natural-language input: ≈ 0 in the symbol-only engine, $+0.07$ – 0.13 across layers 7–15 in the elicit parent ($10\times$). PC1 of the parent’s own answer-position states: 0.23 (elicit parent, problem-specific) vs 0.98 (blank base) and 0.99 (format-installed base) — the same state for every problem (Table 8, “parent’s own states”). *Reading:* the words already reach the elicit parent’s machinery before any target training; the teach parents have nothing for them to reach.

M10 Answer depth — *the answer already there vs. absent; finished one layer earlier vs. later*

Elicit vs. teach, in one sentence. Is the answer readable from the parent’s internal state before it can say it (elicit) or absent (teach); and after training, is it finished one layer earlier (elicit) or later (teach)?

How it is measured. Read the model’s intermediate state at each layer as if it were the output, three ways. *Logit lens:* apply the final unembedding to the layer- ℓ state. *J-lens* (Jacobian lens): first transport the state to the final-layer basis with the average Jacobian $\partial h_{\text{final}}/\partial h_{\ell}$ over held-out story text, then unembed — this reads what the state is *disposed to make the model say*, and is where the answer can appear before the model emits it. *R-lens:* the same transport with a relevance-propagation backward pass, more faithful at early layers. Scored token = the first digit chunk of the answer (for negative answers the sign is fed in first, so predicting a minus sign never counts). Per layer: the answer token’s median rank among 128K vocabulary entries (chance 64K); the *logit-diff*, the model’s preference for the correct answer over a mismatched problem’s answer, in nats (0 = no answer-specific signal); and the *settled depth*, the first layer from which the token stays top-ranked to the end. Rank without logit-diff can be a number prior (any number ranks well in a model that emits numbers); the two are read together.

The numbers. Figure 6, Table 9. Before training, the elicit parent ranks the answer 174 of 128K in its output and 510 at layer 14 (J-lens), preferring it by 2.1 nats while it emits a rewrite; the blank base ranks it $\sim 75\text{K}$ with logit-diff 0.0, and the format-installed base ranks it 452 with logit-diff 0.0 (a number, not this number). After training, elicited children settle at layer 14 (J-rank 338 / 8 / 1 at layers 12 / 13 / 14, full FT); every taught child settles at layer 15 (27,044 / 15,058 / 961 format-installed full FT; 4,123 / 1,832 / 138 blank full FT). *Reading:* the elicit parent already carries the answer and elicitation amplifies the read-out; the blank base carries nothing, and a taught model finishes the answer one layer later even when it solves the task.

M11 Switch-on — *can the capability be turned on without training?*

Elicit vs. teach, in one sentence. Can the task be unlocked in the parent with its weights frozen — possible only if the machinery is already there (elicit) and impossible if it is not (teach)?

How it is measured. Exact match on the target task under interventions that train nothing. (a) *Self-chain:* let the parent rewrite the question in its own symbols, then feed it its own

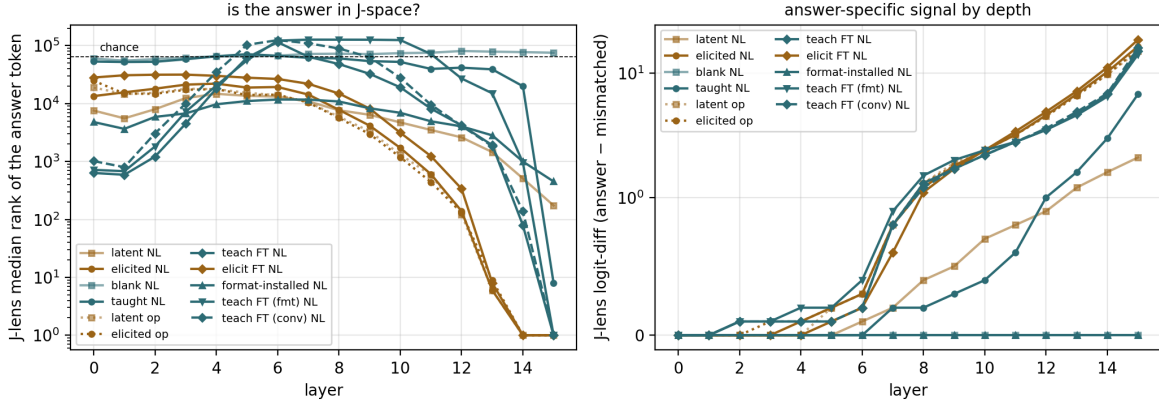


Figure 6: J-lens read-out of the answer token by layer: median rank (left, chance 64K) and logit-diff against a mismatched problem’s answer (right). Squares = parents, triangles = the format-installed base and its child, circles = LoRA children, diamonds = full-FT children; dotted = symbol surface.

model	surface	own acc	rank L12 (logit / J / R)	rank L14 (logit / J / R)	read-out	ld
elicit parent (TS1B-latent)	NL	0.047	5,704 / 2,606 / 2,730	436 / 510 / 424	174	+2.1
blank base	NL	0.000	76,049 / 80,840 / 78,243	77,707 / 77,320 / 79,116	75,441	0.0
format-installed base	NL	0.004	8,394 / 4,036 / 5,417	1,106 / 997 / 998	452	0.0
elicted child (LoRA)	NL	0.957	1,584 / 139 / 464	1 / 1 / 1	1	+16.5
elicted child (full FT)	NL	0.988	1,797 / 338 / 492	1 / 1 / 1	1	+18.7
taught child (full FT, blank, converged)	NL	0.844	18,988 / 4,123 / 9,620	170 / 138 / 459	1	+16.1
taught child (full FT, format-installed)	NL	0.535	23,909 / 27,044 / 25,529	381 / 961 / 1,848	1	+14.2
taught child (LoRA 4M)	NL	0.117	54,896 / 41,722 / 41,666	40,496 / 17,183 / 21,998	6	+7.6
taught child (LoRA 1M)	NL	0.109	51,167 / 41,677 / 44,335	35,620 / 19,933 / 21,156	8	+6.8
elicit parent (TS1B-latent)	symbols	0.836	1,224 / 122 / 334	1 / 1 / 1	1	+15.1
elicted child (LoRA)	symbols	0.855	1,360 / 131 / 399	1 / 1 / 1	1	+15.5

Table 9: Median rank of the answer’s first digit token among 128K entries under the three lenses; “own acc” = the model’s own first-digit-token accuracy on the 256 problems; read-out = the last layer, where all lenses equal the model’s output; ld = logit-diff at the read-out (nats). Blank-base teach FT before continuation: 18,502 / 4,196 / 9,429, then 141 / 79 / 377, own acc 0.797, ld +15.2.

rewrite. (b) *Donor patching*: take the fine-tuned child’s activations at the 32 circuit nodes (M2) and write them into the parent at the answer position, either as one mean vector over problems or as each problem’s own activations (“per-prompt states”); the same with 32 random nodes as control. “Format” = the fraction of outputs that are a number at all. Done for the elicit side (child into the elicit parent) and the teach side (taught child into its own parent: the blank base, or the format-installed base).

The numbers. Figure 7, Table 10. Elicit parent, weights frozen: self-chain **0.328** (= rewrite accuracy $\times$ symbol accuracy, 0.484×0.668); its child’s per-prompt circuit states patched in 0.12 (random nodes 0.000). Teach parents: 0.000 under every dose and chain; taught children’s states patched into their own parent ≤ 0.004 (blank) and 0.016 (format-installed). *Reading*: the latent capability is one composition step or one activation patch from the task; the absent one cannot be switched on because there is nothing underneath.

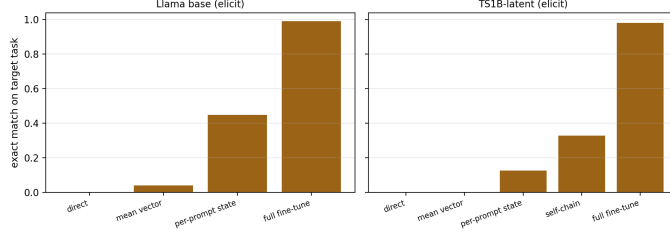


Figure 7: Intervention ladder on the elicit parent (weights frozen except the last bar).

Patient + intervention (32 circuit nodes)	condition	format	exact match
elicit parent + LoRA child’s mean vector	circuit nodes	0.164	0.000
elicit parent + LoRA child’s per-prompt states	circuit nodes	0.359	0.125
	random-32 nodes	0.051	0.000
elicit parent + full-FT child’s per-prompt states	circuit nodes	0.387	0.121
elicit parent self-chain (no donor)	own rewrite + “= ”	–	0.328
blank + binding dose, self-chain	parser without engine	–	0.000
blank + taught donor (LoRA 1M / LoRA 4M / full FT / converged)	every condition	≤0.02	≤0.004
format-installed base + its taught child’s per-prompt states	circuit nodes	0.836 (base 1.000)	0.016

Table 10: Zero-training interventions.

E. Every metric on the full-fine-tune endpoints

Both regimes under one method. The elicit twin and two teach twins, each fully fine-tuned with the identical recipe on the identical 4M file to convergence. The format-installed teach twin is the design-matched comparison (both parents already output numbers); the blank-base teach twin is the better-performing one. They agree on every row.

	elicit FT	teach FT, format-installed	teach FT, blank (conv.)	metric
exact match (0-shot)	0.986	0.424	0.819	M1
steps to stop (passes)	21,500 (0.7)	58,500 (1.9)	72,000 (2.3)	M1
held-out test loss (nats/token)	0.015	0.639	0.154	M1
circuit overlap with the elicit-FT circuit	–	0.306	0.333	M2
circuit overlap with the LoRA-elicited child	0.641	0.333	0.333	M2
top-8 nodes sufficient / necessary	0.923 / 0.943	0.983 / 0.988	0.950 / 0.965	M3
operand-reading heads vs the parent’s engine (first / second)	0.694 / 0.857	0.879 / 0.659	0.763 / 0.658	M4
gradient norm, first 1% → last 10% of steps	4.80 → 1.39 (decays 3.4×)	2.69 → 3.77 (grows 1.4×)	2.30 → 3.43 (grows 1.5×)	M6
cumulative gradient mass	44,021	378,170	353,277 (+ 30,954)	M6
relative weight write $\ \Delta W\ /\ W\ $	0.050	0.086	0.102	M7
PC1 of the state shift (after removing answer content)	0.056 (0.061)	0.567 (0.561)	0.526 (0.528)	M8
answer settled depth, median (quartiles)	layer 14 (14–14)	layer 15 (15–15)	layer 15 (15–15)	M10
J-space rank of the answer at layers 12 / 13 / 14	338 / 8 / 1	27,044 / 15,058 / 961	4,123 / 1,832 / 138	M10
child’s states patched into its own parent, per-prompt (mean)	0.121 (format only)	0.016 (0.000)	0.004 (0.000)	M11

Table 11: The full-FT endpoints, one row per metric.

Notes on measurement

- **Teach-side metrics were re-measured on performing taught models.** The first taught endpoint (LoRA 1M, exact match 0.093) had not learned the task. Every teach-side metric was re-run on the LoRA-4M and the three full-FT taught endpoints, with the full-FT elicited endpoint as the method control. Three earlier readings changed and are stated in their revised form above: taught operand roles are mostly shared rather than absent (M4), the circuit difference is partial rather than total (M2), and the size of the taught state shift

depends on the fine-tuning method (M8).

- **Answer depth is scored on the first digit token.** Scoring the raw first token credited a taught child with layer-5 formation on 48 problems — all subtractions with a negative answer, whose first token is the minus sign, produced by a comparison circuit. With the sign fed in first the effect vanishes; the other models were unaffected.
- **Construction controls.** The same doses on the blank twin change nothing on the target (0.000 in every probe); a sum-only binding dose leaves “difference” unbound; the composition law self-chain = rewrite $\times$ compute held on four variants.